Replication Report: *Microscopic origin of the spin-splitting in altermagnets* (Lee, Kim & Kim, 2025)

Automated physics replication (Hermes subagent)

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1 Headline claim tested

The paper proposes a minimal two-dimensional square-lattice, four-band model (basis $[A\uparrow, B\uparrow, A\downarrow, B\downarrow]^T$) whose spin splitting arises from the interplay of an atomic exchange field $h_{\rm eff}$ and an anisotropic third-neighbour hopping δt generated by distortion of non-magnetic ion cages. The testable headline: the momentum-dependent spin splitting is nonzero *only when both* $h_{\rm eff} \neq 0$ and $\delta t \neq 0$, with a d-wave nodal spin structure.

2 Model (paper Eqs. 1–3)

$$H = \epsilon_k\, I + t_{k,x}\,\tau_x + t_{k,z}\,\tau_z + h_{\rm eff}\,(\tau_z \otimes \sigma_z), \tag{1}$$

$$\epsilon_k = 2t_2(\cos k_x + \cos k_y) + 4t_3 \cos k_x \cos k_y - \mu, \tag{2}$$

$$t_{k,x} = 4t_1 \cos(k_x/2)\cos(k_y/2), \qquad t_{k,z} = -4\,\delta t\,\sin k_x\,\sin k_y, \tag{3}$$

$$\epsilon_{\alpha=\pm,\sigma} = \epsilon_k \pm \sqrt{t_{k,x}^2 + (t_{k,z} + h_{\rm eff}\sigma)^2}. \tag{4}$$

Parameters (energy unit t_1): t_1=1, t_2=0.5, t_3=0.25, δt=0.2, $h_{\rm eff}$=2.0, μ=0.

3 Method

We built the 4×4 Bloch Hamiltonian from scratch and diagonalized it numerically (`numpy.linalg.eigh`), assigning spin labels via $\langle\sigma_z\rangle$. Independent numeric bands were verified against the closed-form Eq. (3), then the spin splitting $\Delta E(\mathbf{k}) = |E_{+,\uparrow} - E_{+,\downarrow}|$ was scanned across parameters and the Brillouin zone.

4 Results

The two-ingredient case study is decisive: with δt=0.2, $h_{\rm eff}$=0 (distortion only) the maximum splitting over the BZ is 0; with δt=0, $h_{\rm eff}$=2 (exchange only) the momentum-dependent splitting is likewise 0 (an isotropic AFM-like gap, no altermagnetic anisotropy); only $\delta t\neq 0$ *and* $h_{\rm eff}\neq 0$ yields the finite anisotropic splitting ($1.174\,t_1$). The splitting generator $t_{k,z} \propto \sin k_x \sin k_y$ is a d_{xy} form factor, giving nodes along k_x=0/k_y=0 and a sign change between BZ quadrants — the hallmark d-wave nodal spin structure.

Check	Result	Status
Numeric 4×4 vs analytic Eq.(3)	max err 1.8×10^{-15}	PASS
Spin splitting at $(\pi/2, \pi/2)$	$\Delta E = 1.109\, t_1$	PASS
ΔE monotonic in δt (Fig. 4d)	$0 \to 2.47$ over $\delta t \in [0, 0.5]$	PASS
ΔE monotonic in h_{eff} (Fig. 4d)	$0 \to 1.43$ over $h \in [0, 4]$	PASS
Anisotropic split requires BOTH terms	only "both_on" nonzero (1.174)	PASS
Nodal along k_x=0 and k_y=0	max split = 0 on both lines	PASS
d-wave sign change between quadrants	sign $t_{k,z}$: $- \to +$	PASS

Table 1: Self-scored checks. 7/7 pass.

5 Verdict

REPLICATED. All seven quantitative checks of the minimal-model headline claim reproduce the paper's stated behaviour. Coverage 8/10 (model core + all qualitative claims replicated; DFT/MnF_2 quantitative fit and SOC/AHE topology out of scope). Agreement 10/10 on the analytic band structure and the two-ingredient necessity rule.

6 Credits

Kubo/Berry methodology reference: `gobel2024_sd_skyrmion_kubo_Lz_kernel.py` (shared-kernels-cache), cited for the anomalous-Hall follow-up; not required for the SOC-free core here. Physics runner: `comfyui-env/bin/python`.